

Greenhouse Gases and Climate Change

Human activities release a variety of gaseous waste products. People once thought that the vast atmosphere could absorb these materials indefinitely, but we now know that our actions can lead to **climate change**, a directional change to the global climate that lasts for three decades or more (as opposed to short-term changes in the weather).

Rising Atmospheric CO₂ Levels

To see how human actions can cause climate change, consider atmospheric CO₂ levels. Over the past 170 years, the concentration of CO₂ in the atmosphere has been increasing as a result of the burning of fossil fuels and deforestation. Scientists estimate that the average CO₂ concentration in the atmosphere before 1850 was about 274 ppm. In 1958, a monitoring station began taking accurate measurements on Hawaii's Mauna Loa peak, a location far from cities and high enough for the atmosphere to be well mixed. At that time, the average CO₂ concentration was 315 ppm (**Figure 56.29**). By 2018, the CO₂ level exceeded 410 ppm, an increase of more than 50% since the mid-19th century. In the **Scientific Skills Exercise**, you can graph and interpret changes in CO₂ concentration that occur over three ten-year periods of time.

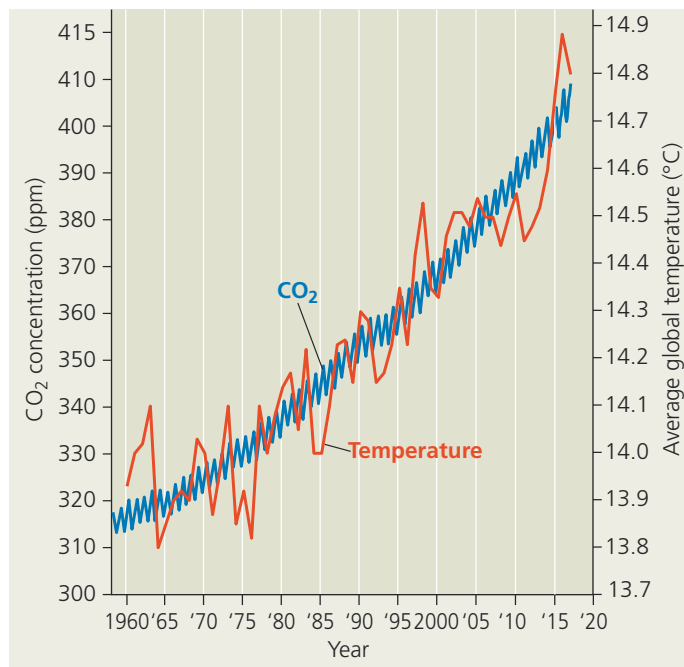
The increase in the concentration of atmospheric CO₂ over the last 170 years concerns scientists because of its link to increased global temperature. Much of the solar radiation that strikes the planet is emitted toward space as infrared radiation (known informally as “heat radiation”).

Although CO₂, methane, water vapor, and other greenhouse gases in the atmosphere are transparent to visible light, they intercept and absorb much of the infrared radiation that Earth emits, radiating most of it back toward Earth. This process, called the **greenhouse effect**, retains some of the solar heat (**Figure 56.30**). If it were not for this greenhouse effect, the average air temperature at Earth's surface would be a frigid -18°C (-0.4°F), and most life as we know it could not exist.

As the concentrations of CO₂ and other greenhouse gases rise, more solar heat is retained, thereby increasing the temperature of our planet. So far, Earth has warmed by an average of 0.9°C (1.6°F) since 1900. Much of this warming has occurred recently: 18 of the 19 warmest years on record have occurred since 2001, with 2016 being the warmest year ever (see **Figure 56.29**).

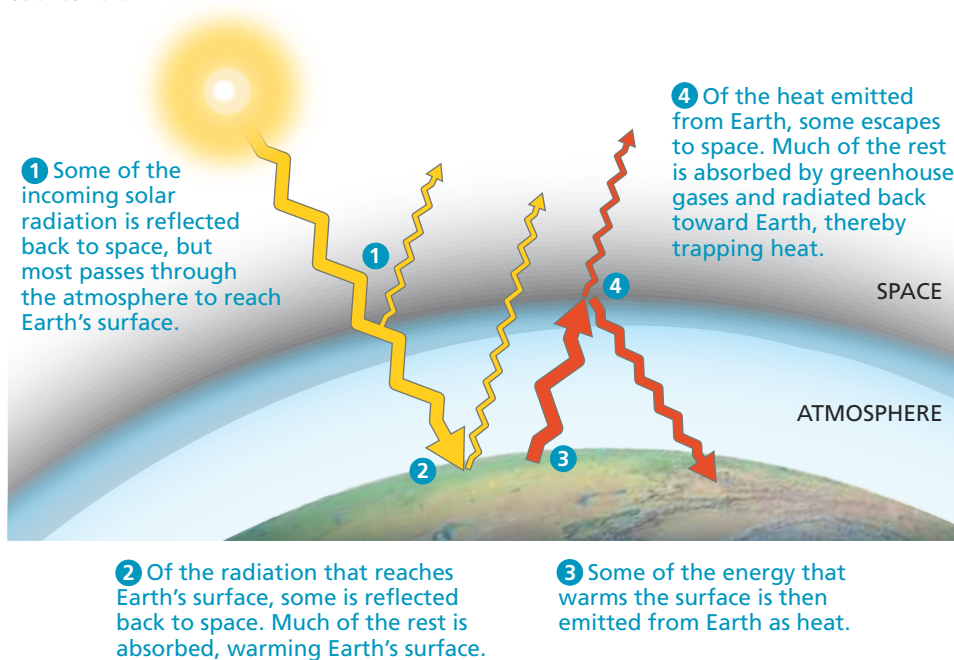
As our planet warms, the climate is changing in other ways as well: Wind and precipitation patterns are shifting,

▼ **Figure 56.29** Increase in atmospheric carbon dioxide concentration at Mauna Loa, Hawaii, and average global temperatures. Aside from normal seasonal fluctuations, the CO₂ concentration (blue curve) increased steadily since 1958. Though average global temperatures (red curve) fluctuated a great deal over time, there is a clear warming trend.



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▼ **Figure 56.30** The greenhouse effect. Carbon dioxide and other greenhouse gases in the atmosphere absorb heat emitted from Earth's surface and then radiate much of that heat back to Earth.



➔ **Mastering Biology** HHMI Animation: Greenhouse Effect
Animation: The Global Carbon Cycle and the Greenhouse Effect

